SimpsonEF: EXPLAINABLE EJECTION FRACTION FROM PREDICTED SONOGRAPHER TRACINGS

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ABSTRACT

Video-based deep learning models accurately estimate left-ventricular ejection fraction (EF), but output a scalar whose clinical derivation cannot be audited. We present SimpsonEF, a framework that replicates the sonographer workflow: it predicts the volume tracings of every beat and derives EF strictly via Simpson's method of disks, so the reported EF is an inspectable geometric function. SimpsonEF uses three lightweight networks: FrameTracer traces all frames, BeatFinder identifies the beats on the predicted volume curve, and BeatTracer jointly refines each beat's end-diastolic and end-systolic tracings. On the EchoNet-Dynamic test set, SimpsonEF achieves an R^2 of 0.804 and an MAE of 4.02 EF points with 4.5M parameters, non-inferior to released video regressors. Because the volume formula is explicit, it adapts to another measurement convention without retraining: on CAMUS, with no network trained on CAMUS data, it reaches $R^2 = 0.710$ against biplane labels, whereas eight released EF models reach at most 0.28. Split-conformal intervals cover 90.3% of test cases, and the predicted tracings segment the left ventricle with a mean Dice of 92.85%. The code is publicly available at `https://github.com/gowthamaanp/simpsonef`.

***Index Terms*—** Ejection fraction, echocardiography, Simpson's method, keypoint detection, interpretability, conformal prediction.

1. INTRODUCTION

Left-ventricular ejection fraction (LVEF, here EF) is a principal metric of systolic function. Video networks estimate EF to within ≈ 4 percentage points [1, 2] but output an uninterpretable scalar. Clinically, EF is derived from anatomical tracings via Simpson's method of disks [3]. A pipeline that outputs the tracing and computes EF only by the formula lets clinicians inspect and correct every step. By contrast, direct regression methods, including 3D CNNs [1], CNN-transformers [4], contrastive models [2], and lightweight spatio-temporal architectures [5], plateau at an R^2 of 0.81–0.82 (EchoNet-Dynamic Test). Similarly, multi-billion-parameter foundation models remain opaque black boxes [6, 7, 8, 9], whereas graph-attention models provide post-hoc explanations while still relying on direct regression [10]. Segmentation models estimate EF from masks [11], though explicit masks do not enhance regressed EF [12]. Keypoint approaches align best with clinical practice: EchoTrace [13] regresses tracing coordinates, and EchoGraphs [14] detects contour keypoints via graph networks. However, EchoGraphs reports that Simpson's EF from its keypoints ($R^2 = 0.73$) trails its own regression head (0.81). To address these limitations, we propose SimpsonEF, an interpretable geometric framework for video-based EF estimation. SimpsonEF predicts the sonographer volume tracings and estimate EF using the Simpson's method of disks.

The main contributions of this work are:

1. An interpretable video-to-EF pipeline driven strictly by Simpson's formula on predicted volume tracings with fully inspectable intermediates.
2. A joint ED/ES tracing architecture closing the MAE gap to direct regression without sacrificing anatomical accuracy.
3. A cross-model benchmark across EchoNet-Dynamic and CAMUS that characterizes formula-derived EF accuracy, zero-shot protocol adaptation and phase detection error.

2. METHODOLOGY

2.1. Tracings and Volume Estimation

Following the EchoNet-Dynamic [1] convention, each landmark tracing consists of the two endpoints of the longitudinal axis, A_1 and A_2, along with the endpoints L_i, R_i of 20 orthogonal chords ($i \in \{1, \ldots, 20\}$). Let $d_i = \|L_i - R_i\|$ denote the chord lengths and $\ell = \|A_1 - A_2\|$ represent the long-axis length. Left Ventricular (LV) volume V and EF are calculated using the method of disks:

$$V = \frac{\pi}{4} \frac{\ell}{20} \sum_{i=1}^{20} d_i^2, \qquad \mathrm{EF} = 100 \left(1 - \frac{V_{\mathrm{ES}}}{V_{\mathrm{ED}}}\right), \tag{1}$$

where V_{ED} and V_{ES} correspond to end-diastolic (ED) and end-systolic (ES) volumes, respectively. Masks are never used for training or EF, only to score Dice.

2.2. Landmark Tracing Networks

Landmark prediction is performed by two fully convolutional encoder–decoder networks, designated as FrameTracer and BeatTracer. Both networks incorporate instance normalization, GELU activations, skip connections, and a dilated bottleneck. Landmarks are extracted via soft-argmax operators [15] evaluated over a 7×7 local window centred on heatmap peaks. Shared weights run at $1\times$, $1/2\times$ and $1/4\times$ resolution with logits fused by log-sum-exp, keeping FrameTracer's VAL Dice within 0.011 for input scales $0.5\times$ to $3\times$. FrameTracer (1.42M parameters) operates on 13-frame temporal sub-sequences to predict 42 landmark coordinates. BeatTracer (2.53M parameters) takes channel-stacked 13-frame windows centred around ED and ES frames as input. It outputs both tracings (84 points), so ES is traced conditioned on ED.

2.3. Cardiac Cycle and Beat Detection

A two-layer bidirectional Gated Recurrent Unit network, BeatFinder (0.53M parameters), processes the temporal chord sequence $x_t = (d_1, \ldots, d_{20}, \ell)$ generated by FrameTracer. BeatFinder outputs continuous ED and ES probability heatmaps across time alongside direct

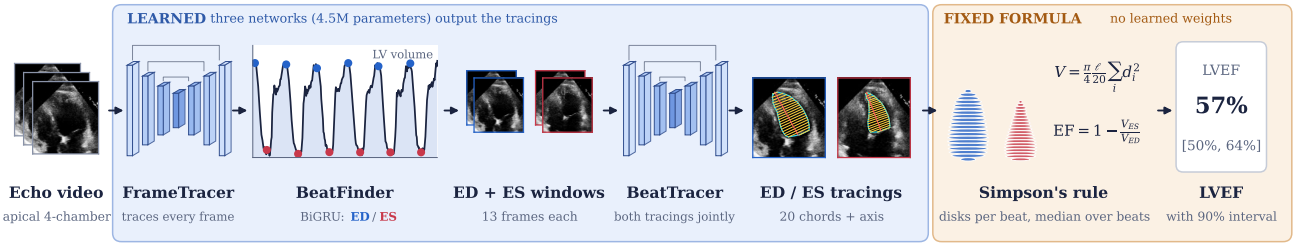


Fig. 1. **SimpsonEF overview.** Learned (blue): FrameTracer traces every frame; BeatFinder finds ED and ES frames from the volume curve; BeatTracer jointly traces ED and ES from local windows. Fixed (orange): disk integration of chords into volumes, and EF as the median over beats with a 90% conformal interval.

EF quantile predictions supervised via pinball loss [16]. Heatmap peaks give ED frames and their following ES frames; without a valid pair, FrameTracer's volume-curve extremes are used.

2.4. Loss Functions and Optimization

Both tracer models are trained using a compound objective function comprising an L_1 point localization loss, a heatmap cross-entropy loss, and L_1 supervision applied to $\log(1+V)$, the relative volume change ($\log V_{ED} - \log V_{ES}$), and log-transformed chord dimensions weighted by their proportional contribution to total volume V. ES frames are weighted 1.25 relative to ED. Augmentation: rotation $\pm 20°$, scale 0.8–1.2, translation ± 18 px, spatial rescaling of the whole canvas by 0.5–2.5×, edge occlusion and synthetic acoustic degradation (FrameTracer only). The relative volume change is $-\log(1-\mathrm{EF})$, so its cost per EF point scales as $1/(1-\mathrm{EF})$ and under-weights low EF. BeatTracer is therefore fine-tuned with an explicit EF loss:

$$\mathscr{L}_{\mathrm{EF}} = w_b \left| \mathrm{EF}(\hat{P}_{\mathrm{ED}}, \hat{P}_{\mathrm{ES}}) - \mathrm{EF}^* \right|, \tag{2}$$

where $\mathrm{EF}(\cdot)$ evaluates Eq. (1) on predicted point sets $\hat{P}$, EF^* is the ground-truth fraction, and w_b denotes class-balancing weights proportional to the inverse square root of training frequencies across binned EF ranges. BeatTracer is trained without synthetic scanner degradation in both stages, to mitigate attenuation bias [17].

2.5. Inference and Conformal Uncertainty Quantification

At inference BeatTracer sees 15 transformed copies of each cardiac cycle (5 scales from 0.85 to 1.17 × rotations of $-8°, 0°, 8°$), whose back-projected points are averaged before Eq. (1) and the median over beats is reported. A normalized split-conformal interval [18, 16, 19] uses the scale

$$s = \frac{\hat{q}_{.95} - \hat{q}_{.05}}{2} + \left| \widehat{\mathrm{EF}} - \widehat{\mathrm{EF}}_{\mathrm{GRU}} \right|, \tag{3}$$

where $\hat{q}_\tau$ represents EF quantiles predicted from chord dynamics by BeatFinder. s grows when the quantiles are wide or the formula and BeatFinder disagree. Given the conformal quantile Q computed over normalized absolute residuals $|y - \widehat{\mathrm{EF}}|/s$ on VAL set, the interval $\widehat{\mathrm{EF}} \pm Qs$ has $\geq 90\%$ marginal coverage under exchangeability. VAL also served model selection, so the guarantee is approximate.

2.6. Clinical Interpretability

The reported EF is derived exclusively from Eq. (1). BeatFinder selects the ED/ES frames, and its EF quantiles act solely to set the interval width. Clinicians can inspect and correct every intermediate: selected frames, tracings, chord lengths, per-beat EF and the causes of a wide interval.

3. EXPERIMENTS AND RESULTS

3.1. Implementation Details

EchoNet-Dynamic [1] contains 7,465 train, 1,288 validation (VAL), and 1,277 test (TEST) videos with EF labels and beat annotations (excluding 5 train and 1 test clip). Training uses 7,460 annotated videos (excluding 81 with stacked annotations). Model selection (tracer variant out of 25 candidates, TTA, beat aggregation) and conformal calibration were performed on VAL only. TEST was evaluated once with frozen weights and no test-time annotations. Metrics: R^2, MAE (EF points), Pearson r, and LV Dice. Tracers are trained from scratch with AdamW (learning rate 3×10^{-4}, weight decay 10^{-4}, cosine decay, EMA 0.999, dropout 0.1, 40 epochs and batch size 32; BeatTracer's EF stage runs 12 epochs (learning rate 10^{-4}, $w_{\mathrm{EF}} = 40$, no dropout). BeatFinder is trained for 16 epochs (learning rate 10^{-3}). Single-video inference takes 0.49 s (0.36 s without TTA) on an NVIDIA RTX 3090.

Model	Year	Par. (M)	Training data	MAE	R^2
EchoNet-Dynamic [1]	2020	31.3	EchoNet-Dynamic	4.01	0.815
EchoGNN [10]	2022	1.7	EchoNet-Dynamic	4.46	0.758
EchoGraphs† [14]	2022	27.6	EchoNet-Dynamic	4.23	0.79
EchoCoTr-B† [4]	2022	50	EchoNet-Dynamic	3.98	0.81
EchoCLIP [20]	2024	151.5	Cedars (0-shot)	11.51	−0.387
CoReEcho† [2]	2024	21	EchoNet-Dynamic	3.90	0.82
Echo-E^{3}Net [5]	2025	1.76	EchoNet-Dynamic	3.92	0.819
EchoFM-L⋆ [8]	2025	344.5	290K clips, probe	5.64	0.627
PanEcho [7]	2025	42.0	Yale (0-shot)	5.47	0.600
EchoJEPA-L⋆ [9]	2026	303.9	MIMIC-IV-Echo, probe	5.48	0.630
EchoPrime [6]	2026	123.1	Cedars (0-shot)	7.78	0.210
SimpsonEF (ours)	2026	4.48	EchoNet-Dynamic	4.02	0.804
EF from keypoints at the annotated ED/ES frames					
EchoGraphs [14]	2022	27.6	EchoNet-Dynamic	4.66	0.73
SimpsonEF (ours)	2026	2.53‡	EchoNet-Dynamic	4.15	0.761

Table 1. EF on EchoNet-Dynamic TEST. †Published figures, others are released weights run locally. ⋆Frozen encoder, probe trained on EchoNet-Dynamic TRAIN. ‡SimpsonEF-BeatTracer.

3.2. Ejection Fraction Estimation

SimpsonEF achieves $R^2 = 0.804$ (95% CI 0.778–0.828), MAE = 4.02, and $r = 0.897$ on EchoNet-Dynamic TEST set (Table 1). Statistical comparisons use only the models run locally under this protocol. Published figures differ in clip sampling or test set and are listed for reference. Under EchoGraphs' keypoint protocol, BeatTracer

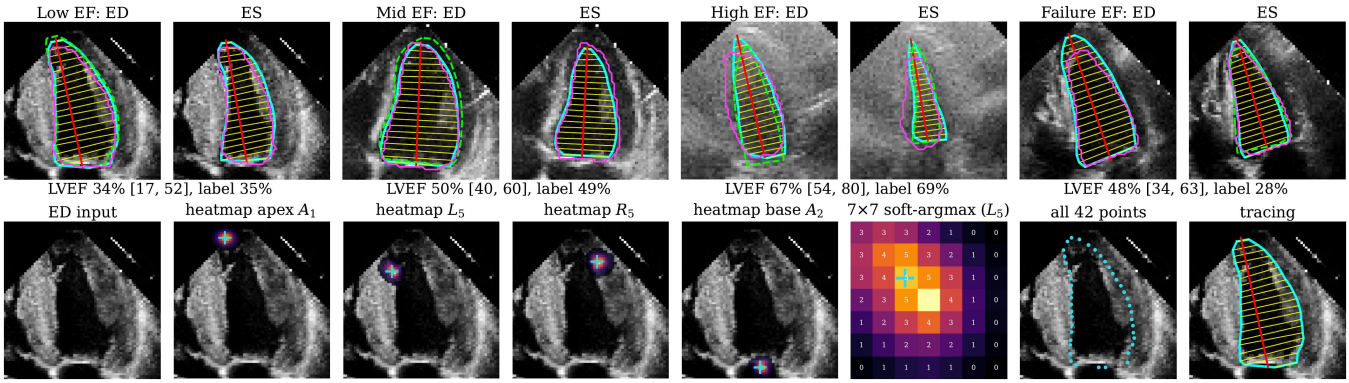


Fig. 2. Top: TEST cases (ED, ES): BeatTracer's tracing (cyan; chords yellow, axis red), EchoNet-Dynamic DeepLabV3 [1] mask (magenta), reference (green dashed); formula EF, 90% interval, label. Bottom: first case, ED: per-point heatmaps (4 of 42), 7×7 soft-argmax at the peak (weights in %) giving the sub-pixel point (+), and the tracing.

reaches R^2 0.761 vs. 0.73 [14]. Paired bootstrap MAE differences (ours minus baseline, 95% CI) show non-inferiority at a 0.3-point margin, not superiority, to EchoNet-Dynamic [1] (+0.01 [−0.16, 0.18]) and Echo-E^{3}Net [5] (+0.10 [−0.07, 0.27]). SimpsonEF beats EchoGNN [10] (−0.44 [−0.63, −0.25]) and the foundation-model probes [9, 8] ($\geq$ 1.46 MAE improvement). EF $<$ 40% is detected with AUROC 0.978 (EchoNet-Dynamic 0.9818, Echo-E^{3}Net 0.9817; sensitivity 71.9% vs. 70.6% and 78.8% at specificity 98.3% vs. 98.1% and 98.2%), and EF $<$ 50% with 0.955.

Model	Year	Par. (M)	Training data	ED	ES	Mean
EchoNet-Dynamic† [1]	2020	39.6	EchoNet-Dynamic	92.3	90.2	91.3
nnU-Net [21]	2021	7.37	EchoNet-Dynamic	93.6	91.6	92.6
SepXception [22]	2022	55.8	EchoNet-Dynamic	93.6	91.7	92.7
SAM†‡ [23] (fine-tuned)	2023	93.7	SA-1B	92.9	89.4	91.1
SimLVSeg [24]	2024	18.8	EchoNet-Dynamic	94.0	92.3	93.1
Echo-ODE [25]	2025	–	EchoNet-Dynamic	93.4	91.5	92.5
SimpsonEF (ours)	2026	2.53⋆	EchoNet-Dynamic	93.7	92.0	92.8

Table 2. LV Dice (%) on EchoNet-Dynamic TEST, annotated ED/ES frames. †Evaluated locally and others are published figures. ‡Fine-tuned on EchoNet-Dynamic TRAIN, ground-truth box prompt. ⋆SimpsonEF-BeatTracer.

Model	Year	Par. (M)	ED	ES	Pool	Acc (%)
UVT† [26]	2021	346.8	7.17	3.36	5.27	–
CardiacPhase [27]	2025	–	3.0	2.0	2.5	–
VidFuncta [28]	2026	–	3.06	2.90	2.98	–
Volume-curve extremes⋆	2026	1.42	2.99	2.70	2.85	57.1
SimpsonEF (ours)	2026	1.95	1.74	1.82	1.78	76.4

Table 3. ED/ES frame error and accuracy (% $\leq$ 2 frames), TEST. All rows: whole video, nearest detection to the annotated frame. †Excludes videos with $|\text{ED}-\text{ES}| \leq 10$ frames and 6 rejected. ⋆SimpsonEF-FrameTracer volume extrema.

3.3. Left Ventricular Segmentation

BeatTracer reaches ED/ES Dice of 93.7% and 92.0% (Table 2), above the EchoNet-Dynamic DeepLabV3 and box-prompted fine-tuned SAM, and approaching mask-trained segmentation models although it predicts keypoints, not masks. Averaged per image instead, it is 93.5% and 91.5% (DeepLabV3 92.0%, 89.7%) and against the released PNG masks (all 42 points), 91.7% and 90.7% on TEST (91.6%, 90.7% on VAL), the convention of Sec. 3.7.

3.4. Cardiac Phase Detection

Supervised by one annotated beat per video, BeatFinder detects ED/ES frames with 1.78 frames mean error (76.4% within ±2 frames; Table 3), measured from each annotated frame to the nearest detected frame of the same phase, the protocol of [26, 27, 28]. BeatFinder fires about once per beat (median 43 frames between detections), so the matching is not lenient. This is lower than UVT and annotation-free methods [27, 28] and than FrameTracer's volume extrema (2.85). Not comparable, hence not tabulated: EchoGNN [10] (3.68/4.15; block-boundary distance on 1,173 videos) and a supervised R3D-50 (2.1/1.7, cited in [27]).

3.5. Cross-Dataset Generalization on CAMUS

CAMUS [29] shifts the protocol: biplane Simpson labels, single half-cycles ($\sim$ 20 frames) and different contour boundaries. As in the EF code released with the dataset, its label uses one disk height for ED and ES (the longest of the four long axes), so with 4- and 2-chamber chord lengths a_i, b_i:

$$\text{EF} = 100\left(1 - \textstyle\sum_i a_i b_i\big|_{\text{ES}} \big/ \sum_i a_i b_i\big|_{\text{ED}}\right).$$

On the reference masks this reproduces the label ($R^2 = 0.998$), whereas per-phase heights [1] give 0.607 (bias +6.9). No CAMUS data were used to train network weights. On the 50 test patients, FrameTracer yields R^2 =0.710 (95% CI 0.49–0.84, MAE = 5.08, $r = 0.849$) and BeatTracer 0.515 (MAE 6.62), with 4-chamber Dice 90.5/86.8% for FrameTracer (Table 4). End-to-end models fail ($R^2 \leq 0.283$; Table 4a), although the comparison favours a formula, since the label is itself a Simpson EF. For 4-chamber segmentation, SimpsonEF (88.7% mean Dice) beats the zero-shot EchoNet-Dynamic DeepLabV3 (84.0%) and box-prompted zero-shot SAM (86.7%), but trails the other models in Table 4(b) by 0.9–5.2 points.

3.6. Error Analysis and Uncertainty Quantification

On the 1,264 TEST videos with a single tracing per annotated frame, Simpson's rule on the sonographer tracings reproduces the label ($R^2 = 0.991$; 0.943 including the 12 with stacked tracings). BeatTracer's tracings give $R^2 = 0.790$, 0.969 with reference chord widths and 0.805 with reference long axes indicating that chord widths are the main error source. Errors compress toward normal

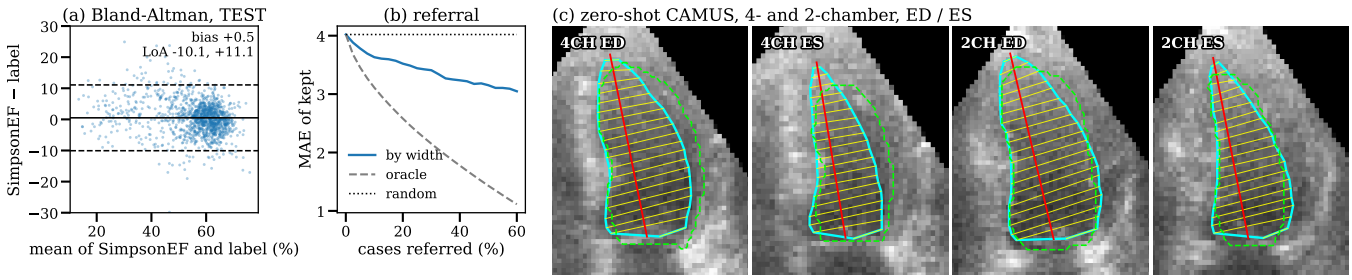


Fig. 3. (a) TEST set Bland–Altman analysis comparing SimpsonEF against clinical ground-truth labels. (b) Trajectory of MAE as a function of uncertainty filtering, excluding the widest conformal intervals. (c) Zero-shot cross-dataset evaluation on CAMUS: FrameTracer's predicted tracing (cyan) versus ground-truth reference contour (green dashed).

(a) EF, biplane label

Model	MAE	R^2
EchoNet-Dynamic [1]	9.71	0.079
Echo-E^{3}Net [5]	11.21	−0.303
EchoGNN [10]	9.94	−0.178
PanEcho [7]	7.53	0.283
EchoPrime [6]	18.63	−1.832
EchoCLIP [20]	9.33	0.072
EchoFM-L [8] (probe)	9.10	0.100
EchoJEPA-L [9] (probe)	10.57	−0.099
SimpsonEF, BeatTracer	6.62	0.515
SimpsonEF, FrameTracer	5.08	0.710

(b) 4CH LV Dice (%)

Model	ED	ES	Mean
EchoNet-Dynamic† [1]	88.1	79.9	84.0
SimLVSeg [24]	91.0	88.2	89.6
SAM, zero-shot†* [23]	87.3	86.1	86.7
SAM, fine-tuned†* [23]	90.4	88.9	89.7
MedSAM2* [30]	94.6	93.1	93.9
DPS-Net [31]	93.2	92.8	93.0
SimpsonEF, FrameTracer†	90.5	86.8	88.7
Run on all 500 videos, full CAMUS dataset			
CAFEx [32]	-	-	90.7

Table 4. CAMUS test set (50 patients), no CAMUS training of our weights. (a) EF: ours biplane Simpson; others, run locally, mean of their 4- and 2-chamber EF on the half-sequence mirrored into an ED–ES–ED cycle. (b) †Run locally and others are published figures. *Ground-truth box prompt.

EF (slope 0.81, Bland–Altman bias +0.5, limits −10.1 to +11.1; Fig. 3a), overestimating by +3.4 to +5.5 points for EF $<$ 50% and underestimating by 5.7 for EF $>$ 70%. Split-conformal intervals cover 90.3% of TEST (95% Wilson CI 88.5–91.8%, mean width 16.7) and 87.5% for EF $<$ 40%. Interval width flags errors $>$ 10 points with AUROC 0.73 and discarding the widest 20% lowers MAE from 4.02 to 3.52 (Fig. 3b).

3.7. Ablation Studies

Table 5 lists the gains on VAL (differences below $\sim$ 0.01 are within seed variation). Joint ED/ES tracing gives the largest step (+0.021 R^2, −0.20 MAE), then the EF-unit loss (+0.016, −0.11) and aggregating beats by the median rather than the mean is worth 0.029. The EF loss does not degrade the anatomy: on VAL without input-copy averaging, Dice (91.6/90.6% before, 91.4/90.5% after) and absolute volume errors (ED 13.2% → 12.7%, ES 16.2% → 15.4%) remain stable or improve. ED and ES volume errors are strongly correlated (0.79 before, 0.80 after), a shared scale error that cancels in the ratio.

4. CONCLUSION

In this work, we introduced SimpsonEF, an interpretable geometric framework demonstrating that Simpson's method of disks applied to predicted anatomical tracings can match the predictive performance of black-box video regressors using a compact parameter footprint of only 4.5M parameters. By grounding the reported ejection fraction strictly in explicit geometric computations, the proposed model provides full clinical inspectability. Furthermore, because the model outputs geometric primitives rather than unconstrained scalar predictions, it adapts zero-shot to alternative clinical measurement protocols such as biplane Simpson labels on external datasets where end-to-end regression models fail to transfer.

Variant	R^2	MAE	r	slope
FrameTracer only	0.776	4.26	0.883	0.81
+ Joint ED/ES tracing (BeatTracer)	0.797	4.06	0.894	0.78
+ EF-unit formula loss	0.813	3.95	0.904	0.82
+ Width 32 expansion	0.820	3.89	0.906	0.83
+ Input-copy averaging (SimpsonEF)	0.820	3.86	0.906	0.83
Single-Modification Ablations				
BeatFinder instead of Simpson formula	0.810	3.87	0.900	0.81
Mean instead of median over beats	0.791	4.08	0.890	0.82
Without EF-unit loss	0.810	4.00	0.900	0.79
25-frame windows, 3 EF stages	0.821	3.89	0.907	0.82
Tail-oversampled EF stage	0.801	4.13	0.899	0.87

Table 5. Ablation study on VAL (1,288 videos, full pipeline, beat selection via BeatFinder). Final model selected on these values (TEST R^2 0.804).

Limitations and Future Work. Despite its strong performance, our approach has limitations. First, residual volumetric errors exhibit variance compression toward normal EF ranges, leading to systematic overestimation at very low EF values ($<$ 40%) and underestimation at high EF values ($>$ 70%). Second, zero-shot cross-dataset evaluation was evaluated on 50 CAMUS test patients. A broader multi-center validation across heterogeneous scanner vendors and clinical acquisition protocols remains an important future direction. Third, split-conformal uncertainty bounds were calibrated under exchangeability, and performance guarantees under severe domain shift require further exploration. Finally, while the system is designed for complete inspectability, formal clinical reader studies remain to be conducted.

5. COMPLIANCE WITH ETHICAL STANDARDS

This study used open-access human subject data from EchoNet-Dynamic and CAMUS. Ethical approval was not required as confirmed by the licences attached to the data.

6. ACKNOWLEDGMENTS

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